



# Modelling Financial Markets

## Chapter 8: Backtesting and Evaluating Risk Forecasts



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## Acronyms used in this material (1/2)

- ACI: Adaptive Conformal Inference
- AI: Artificial Intelligence
- AR: AutoRegressive (model)
- BCBS: Basel Committee on Banking Supervision
- BET: Bucharest Exchange Trading (index)
- BH: Benjamini–Hochberg (procedure)
- BIS: Bank for International Settlements
- BNR: Banca Națională a României (the National Bank of Romania)
- COVID-19: COronaVirus Disease 2019
- DGP: Data-Generating Process
- DM: Diebold–Mariano (test)
- DQ: Dynamic Quantile (test)
- EGARCH: Exponential GARCH
- ES: Expected Shortfall
- ESR: Expected Shortfall Regression (backtest)
- EVT: Extreme Value Theory
- FDR: False Discovery Rate
- FHS: Filtered Historical Simulation
- FRTB: Fundamental Review of the Trading Book
- FWER: Family-Wise Error Rate
- FZ: Fissler–Ziegel (family of scoring functions for VaR and ES)
- FZ0: Fissler–Ziegel loss, zero-homogeneous version
- GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- GARCH-EVT: GARCH filter combined with Extreme Value Theory for the tail
- GAS: Generalised Autoregressive Score (model)
- GJR-GARCH: Glosten–Jagannathan–Runkle GARCH

## Acronyms used in this material (2/2)

- ▣ GPD: Generalised Pareto Distribution
- ▣ HS: Historical Simulation
- ▣ IMA: Internal Models Approach
- ▣ LLM: Large Language Model
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ PIT: Probability Integral Transform
- ▣ POF: Proportion Of Failures (test)
- ▣ POT: Peaks Over Threshold
- ▣ QLIKE: Quasi-LIKElihood (loss function)

## Chapter Route and Outcomes

### Three questions

- ▣ **Calibration:** is a VaR or ES forecast right, day after day, and can a backtest tell?
- ▣ **Comparison:** which of several forecasts is better, and is the gap significant?
- ▣ **Decision:** what can we conclude when the tail holds only a few days a year?
- ▣ Part I: setup, VaR tests, their size and power, regulation, first ES tests; Part II: ES evidence, scores, comparative backtests, conformal VaR, case study, AI mini-case
- ▣ Technical detail sits in the Appendix at the end

### After this chapter you can

- ▣ test the number and the timing of VaR breaches and state the size and power of each test at your  $T$
- ▣ backtest ES with residuals,  $Z_2$  by simulation and PIT-based statistics
- ▣ rank (VaR, ES) forecasts with a consistent score, DM tests and the MCS
- ▣ judge what a distribution-free (conformal) VaR does and does not guarantee

## Setup and Notation

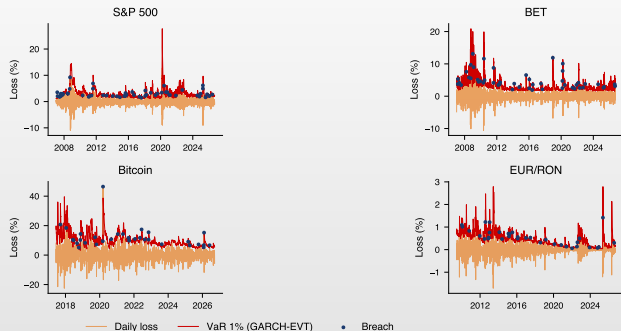
- ▣ **Backtesting**: compare risk forecasts made with the information  $\mathcal{F}_{t-1}$  with the outcomes that followed
- ▣ Daily log return  $X_t = r_t = \ln(P_t/P_{t-1})$ ; daily **loss**  $L_t = -r_t$  (positive number = money lost)
- ▣ **VaR** (Value-at-Risk) and **ES** at tail probability  $\alpha$  (Chapter 7)
  - ▶  $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$ ;  $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du = E[L_t \mid L_t > \text{VaR}_\alpha]$   
(continuous case)
  - ▶ Levels are tail probabilities: VaR 1% ( $\alpha = 0.01$ ), VaR 2.5% and ES 2.5% ( $\alpha = 0.025$ ); Basel used VaR 1% over 10 days, FRTB uses ES 2.5%, backtests use one-day forecasts
- ▣ A **breach** (exception) occurs when the loss exceeds the forecast:  $I_t = 1\{L_t > \text{VaR}_t\}$ , with target rate  $\alpha$
- ▣ Ideal **conditional calibration**:  $P(I_t = 1 \mid \mathcal{F}_{t-1}) = \alpha$  for all  $t$ , so the  $I_t$  are i.i.d. Bernoulli( $\alpha$ ) (Christoffersen (1998))
  - ▶ **Unconditional coverage**: about  $\alpha T$  breaches in  $T$  days; **independence**: breaches do not cluster

## Six Competing Forecasters and the Experiment

| Model     | One-day VaR and ES from                                                                                             |
|-----------|---------------------------------------------------------------------------------------------------------------------|
| HS        | the empirical quantile of order $1 - \alpha$ of the last 1000 losses                                                |
| Normal    | the Normal distribution with the window mean and standard deviation                                                 |
| Student-t | a Student-t fitted to the window (location, scale, $\nu$ )                                                          |
| GARCH-t   | $\sigma_t^2 = \omega + a \varepsilon_{t-1}^2 + b \sigma_{t-1}^2$ , standardised Student-t shocks                    |
| FHS       | GARCH volatility $\times$ empirical quantiles of $z_t = \varepsilon_t / \sigma_t$                                   |
| GARCH-EVT | GARCH volatility $\times$ a GPD tail fitted by POT to the largest 10% of standardised losses (McNeil & Frey (2000)) |

- Four series, each on its own calendar
  - ▶ S&P 500 and BET (Bucharest Stock Exchange index): closes
  - ▶ Bitcoin (trades every day); EUR/RON: BNR reference rate
- Rolling window of 1000 observations; each forecast uses data up to the previous day only
- GARCH, Student-t and EVT parameters re-estimated every 20 days; volatility updated daily
- Evaluation days  $T$ , until 18.09.2026
  - ▶ S&P 500 4 959 and BET 4 937 (from 2007), EUR/RON 4 352 (from 2009), Bitcoin 3 384 (from 2017)

## Four Markets, One Risk Model

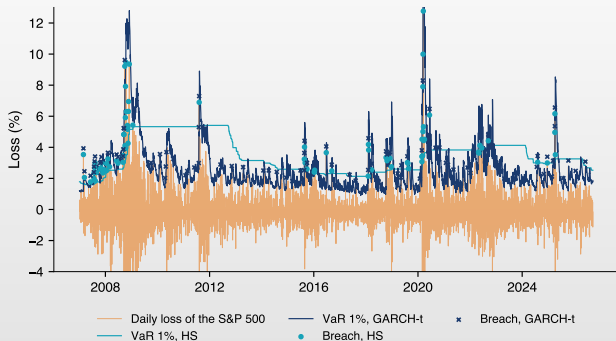


- Orange: daily losses; red: VaR 1% forecast by GARCH-EVT; blue dots: breaches ( $L_t > VaR_t$ )
- The VaR moves with volatility: it doubles or triples in 2008, March 2020 and the 2022 crypto crash
- Breach rates: S&P 500 1.13%, BET 0.99%, Bitcoin 0.95%, EUR/RON 0.78%
- EUR/RON has a much smaller scale: a managed exchange rate, long calm spells, rare jumps
- Question for the chapter: is a rate close to 1% enough to call a model good?

## What a Backtest Can and Cannot Identify

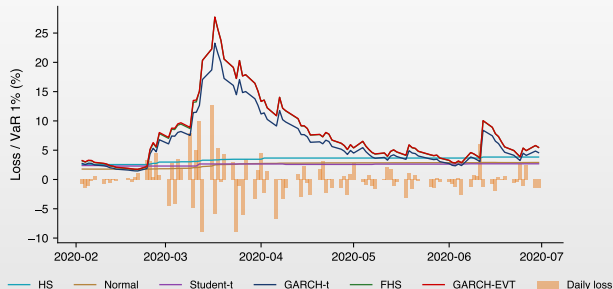
- ▣ Each backtest checks one implication of conditional calibration: Kupiec the rate, Christoffersen first-order memory, durations the spacing, DQ predictability from chosen regressors
- ▣  $H_0$  is a **joint** hypothesis on the whole forecasting procedure
  - ▶ distribution, volatility dynamics, estimated parameters, window length and refit rule
  - ▶ A rejection does not say which component failed
- ▣ One outcome per day: evidence accumulates only over many days, and the textbook  $\chi^2$  limits treat  $\text{VaR}_t$  as known (estimation risk, end of Section 2)
- ▣ Non-rejection is not validation: it is informative only if the test has power against the relevant alternatives at the available  $T$

## S&P 500: Two VaR Forecasts, 2007–2026



- VaR 1%: HS in teal, GARCH-t in blue; orange: daily losses; markers: breaches of each model
- HS is a step function: it jumps only when extreme days enter or leave the 1000-day window; GARCH-t rises the day after volatility rises
- Almost the same count: HS 83, GARCH-t 88 breaches in 4 959 days
- Similar counts can hide different timing: HS breaches arrive in bursts (2008, 2020), so we must also test **when** breaches occur

## March 2020: The Stress Test Nobody Planned

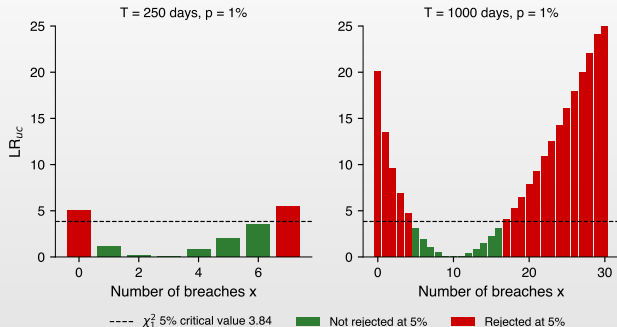


- S&P 500, February–June 2020: daily losses (bars) and the VaR 1% of the six models
- Largest loss 12.77% on 16.03.2020; a correct VaR 1% expects 1.04 breaches in these 104 days
- Breaches observed
  - ▶ unconditional: HS 12, Normal 17, Student-t 17
  - ▶ conditional: GARCH-t 3, FHS 2, GARCH-EVT 2
- In this episode a fat tail without a moving scale (Student-t) did not help: conditioning on current volatility mattered more than the tail shape

## Kupiec (1995): the POF Test

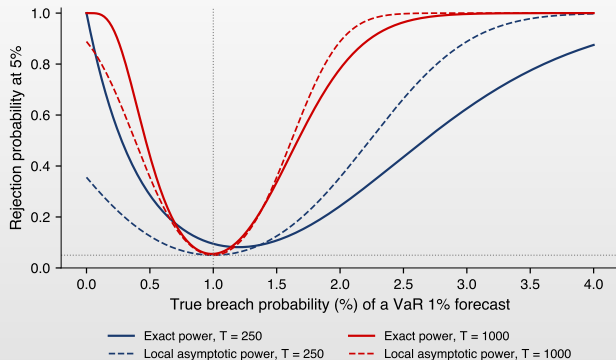
- ▣ Data:  $x = \sum_t I_t$  breaches in  $T$  days; estimated breach rate  $\hat{\pi} = x/T$
- ▣  $H_0: \pi = \alpha$  (correct unconditional coverage), against  $H_1: \pi \neq \alpha$
- ▣ Likelihood ratio statistic (Kupiec (1995))
  - ▶  $LR_{uc} = -2 \ln \frac{(1-\alpha)^{T-x} \alpha^x}{(1-\hat{\pi})^{T-x} \hat{\pi}^x} \xrightarrow{d} \chi_1^2$
- ▣ Reject at 5% if  $LR_{uc} > 3.84$ ; two-sided: too few breaches also count as failure; for small  $T$  use the exact Binomial( $T, \alpha$ ) distribution of  $x$
- ▣ Worked example, S&P 500, GARCH-t, 2008:  $x = 8$  in  $T = 253$  (expected 2.53):  $LR_{uc} = 7.60$ ,  $p = 0.0058$ ; exact Binomial  $p = 2P(X \geq 8) = 0.009$

## How Many Breaches Are Acceptable?



- $LR_{uc}$  against the number of breaches  $x$ , one year (left) and four years (right) of VaR 1%
- $T = 250$ : the test accepts  $x$  from 1 to 6;  $T = 1000$ : from 5 to 16
- Even  $x = 0$  is rejected in one year ( $LR_{uc} = -500 \ln 0.99 = 5.03$ ): a VaR that is never breached is too conservative
- One year holds about 2.5 informative days for VaR 1%: use long samples, several levels ( $\alpha = 1\%$  and  $2.5\%$ ) and several tests

## Power of the POF Test

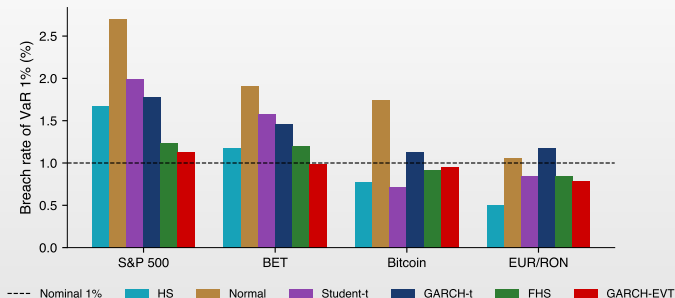


- Rejection probability at 5% against the true breach probability; solid: exact Binomial; dashed: local  $\chi^2_1(\lambda)$
- Local alternatives  $\pi_T = \alpha + h/\sqrt{T}$ :  

$$LR_{uc} = \frac{T(\hat{\pi} - \alpha)^2}{\alpha(1-\alpha)} + o_p(1) \xrightarrow{d} \chi^2_1(\lambda),$$

$$\lambda = \frac{T(\pi - \alpha)^2}{\alpha(1-\alpha)}$$
- $T = 250$ ,  $\pi = 2\%$ :  $\lambda = 2.53$ ; local power 36%, exact power 24%: a doubled breach rate passes three years in four
- The rule “reject if  $LR_{uc} > 3.84$ ” has actual size 9.5% at  $T = 250$  ( $P(x = 0) = 8.1\%$  alone): the size of the  $\chi^2$ -calibrated LR rule, not of an exact 5% test
- 50% power needs  $\pi \approx 2.7\%$  at  $T = 250$  and  $\pi \approx 1.7\%$  at  $T = 1000$

## Breach Rates Across Models and Markets

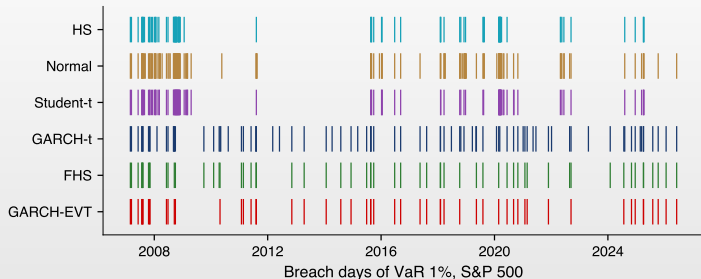


- Share of days with  $L_t > \text{VaR}_t$  for VaR 1%; the Normal distribution fails except on EUR/RON: S&P 500 2.70%, BET 1.90%, Bitcoin 1.74%
- Same GARCH-t volatility, different tail: GARCH-t 1.77% and 1.46% versus FHS 1.23% and 1.20% (S&P 500, BET): evidence against the symmetric t tail on the loss side
- HS is too conservative on EUR/RON (0.51%): an old crisis in the window keeps its VaR high

## Christoffersen (1998): Testing Independence

- ▣ Model  $I_t$  as a first-order Markov chain with  $\pi_{ij} = P(I_t = j \mid I_{t-1} = i)$
- ▣ Count transitions  $n_{ij}$  (from state  $i$  yesterday to state  $j$  today);  $\hat{\pi}_{01} = \frac{n_{01}}{n_{00} + n_{01}}$ ,  $\hat{\pi}_{11} = \frac{n_{11}}{n_{10} + n_{11}}$
- ▣  $H_0$ :  $\pi_{01} = \pi_{11} = \pi$  (no memory), with  $\hat{\pi} = (n_{01} + n_{11}) / \sum n_{ij}$
- ▣ Statistic
  - ▶ 
$$\text{LR}_{ind} = -2 \ln \frac{(1 - \hat{\pi})^{n_{00} + n_{10}} \hat{\pi}^{n_{01} + n_{11}}}{(1 - \hat{\pi}_{01})^{n_{00}} \hat{\pi}_{01}^{n_{01}} (1 - \hat{\pi}_{11})^{n_{10}} \hat{\pi}_{11}^{n_{11}}} \xrightarrow{d} \chi_1^2$$
- ▣ Conditional coverage:  $\text{LR}_{cc} = \text{LR}_{uc} + \text{LR}_{ind} \xrightarrow{d} \chi_2^2$  tests both properties at once

## When Do Breaches Happen?

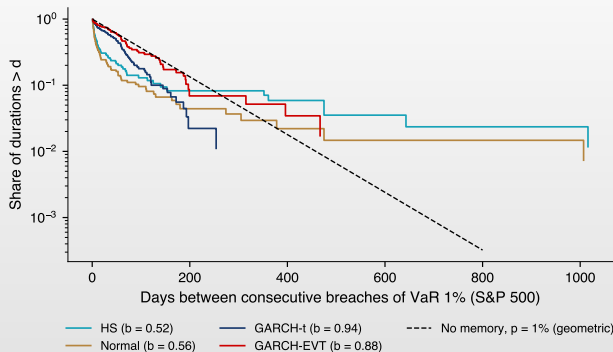


- Each tick: one breach of VaR 1% on the S&P 500; HS, Normal, Student-t: bursts in 2008, 2015–2016, 2018, 2020, 2022; filtered models: spread almost uniformly
- Counterexample, HS in 2007–2010 only:  $n_{00} = 928$ ,  $n_{01} = 38$ ,  $n_{10} = 38$ ,  $n_{11} = 3$ ;  $\hat{\pi}_{11} = 0.0732$  versus  $\hat{\pi}_{01} = 0.0393$ , yet  $LR_{ind} = 0.95$ ,  $p = 0.33$ :  $\hat{\pi}_{11}$  rests on 3 pairs
- Full sample: independence rejected for HS ( $p < 0.001$ ), Normal ( $p < 0.001$ ) and, narrowly, for FHS ( $p = 0.046$ ), not for GARCH-t ( $p = 0.095$ ): a subsample non-rejection is not a full-sample verdict

## Duration-Based Tests

- ▣ Duration  $D_i$ : number of days between breach  $i - 1$  and breach  $i$
- ▣ Under a correct VaR, breaches have no memory:  $D_i$  is geometric with mean  $1/\alpha$  (100 days for  $\alpha = 1\%$ )
- ▣ Christoffersen & Pelletier (2004): approximate the discrete durations by a continuous Weibull model
  - ▶  $f(D; a, b) = a^b b D^{b-1} e^{-(aD)^b}$ ;  $b = 1$  is the exponential (memoryless) case
  - ▶  $b < 1$ : short durations are too frequent, i.e. breaches cluster
- ▣  $H_0$ :  $b = 1$  with  $a$  free, a likelihood ratio test ( $\chi^2_1$ ) of memorylessness, not of the mean  $1/\alpha$  (coverage is left to Kupiec); first and last durations censored
- ▣ Uses the whole spacing pattern, not only consecutive days: more power against clustering

## How Long Between Breaches?

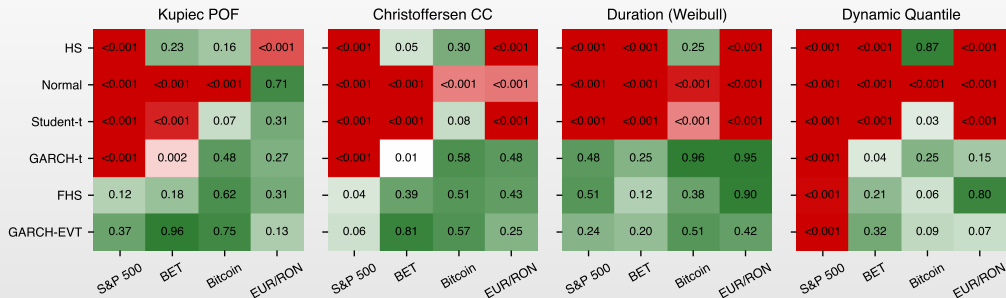


- Share of durations longer than  $d$  (log scale); dashed: no memory with  $\alpha = 1\%$
- HS:  $b = 0.52$ , median 7 days, 44% of durations at most 5 days; test  $p < 0.001$
- Normal:  $b = 0.56$ ; GARCH-t:  $b = 0.94$  (median 48 days,  $p = 0.479$ ); GARCH-EVT:  $b = 0.88$  ( $p = 0.240$ )
- The long tail of HS durations: years without breaches after a crisis leaves a high VaR in the window

## Regression-Based Backtests: the DQ Test

- Under correct conditional coverage  $\text{Hit}_t = I_t - \alpha$  is a martingale difference:  $E[\text{Hit}_t \mid \mathcal{F}_{t-1}] = 0$ ,  $\text{Var}(\text{Hit}_t \mid \mathcal{F}_{t-1}) = \alpha(1 - \alpha)$
- So no  $\mathcal{F}_{t-1}$ -measurable variable may predict it: regress  $\text{Hit}_t = X_t\beta + u_t$  with  $X_t = (1, \text{Hit}_{t-1}, \dots, \text{Hit}_{t-4}, \text{VaR}_t)$  and test  $\beta = 0$
- Dynamic Quantile statistic ([Engle & Manganelli \(2004\)](#)), Wald form with the null variance)
  - ▶  $\text{DQ} = \frac{\hat{\beta}' X' X \hat{\beta}}{\alpha(1 - \alpha)} = \frac{\text{Hit}' X (X' X)^{-1} X' \text{Hit}}{\alpha(1 - \alpha)} \xrightarrow{d} \chi_6^2$
- Nests Kupiec ( $X_t = 1$  gives the score form  $T(\hat{\pi} - \alpha)^2 / (\alpha(1 - \alpha))$ ) and Christoffersen (one lag); adds power against memory beyond lag 1 and against hits that depend on the VaR level

## All VaR Tests at a Glance



- $p$ -values of the Kupiec, conditional coverage, duration and DQ tests, VaR 1%; red: rejection, green: no rejection
- DQ rejects every S&P 500 model, including FHS ( $p < 0.001$ ) and GARCH-EVT ( $p < 0.001$ ): hits are predictable from the VaR level and lagged hits, a failure that counts miss
- Different tests examine different failures: the unconditional models fail durations almost everywhere; GARCH-EVT passes coverage and durations on BET, Bitcoin and EUR/RON; read the pattern, not one  $p$ -value

## Size and Power of VaR Backtests: a Monte Carlo

- DGP: GARCH(1,1)-t fitted to the S&P 500 1990–2026 ( $a = 0.102$ ,  $b = 0.892$ ,  $\nu = 6.1$ ); VaR 1%; 1000 paths; rejection rates (%) at 5%

| Forecaster            | $T$  | POF         | CC   | Duration    | DQ         |
|-----------------------|------|-------------|------|-------------|------------|
| Correct (size)        | 250  | <b>9.8</b>  | 1.2  | 14.8        | 8.6        |
|                       | 1000 | 6.5         | 3.3  | 7.9         | 11.0       |
|                       | 5000 | 4.9         | 3.4  | 4.6         | <b>8.8</b> |
| Normal-GARCH (power)  | 250  | <b>10.5</b> | 4.9  | 14.7        | 18.8       |
|                       | 1000 | 30.3        | 19.9 | 8.0         | 30.3       |
|                       | 5000 | 83.8        | 77.2 | 7.2         | 70.7       |
| HS, 1000 days (power) | 250  | 51.2        | 17.2 | <b>12.3</b> | 32.4       |
|                       | 1000 | 46.3        | 47.1 | 68.5        | 73.8       |
|                       | 5000 | 57.1        | 86.4 | 100.0       | 99.5       |

- Correct: true parameters and  $\sigma_t$ ; Normal-GARCH: true  $\sigma_t$ , Normal tail; an undefined duration  $p$  (fewer than 3 durations, 28% of correct samples at  $T = 250$ ) counts as non-rejection; Monte Carlo s.e. 0.7 points at 5%, 1.6 at 50%
- In bold: in one year a Normal tail on the right volatility is almost invisible and HS clustering is seldom detected; DQ stays oversized even at  $T = 5000$ , so simulate its critical values

## Backtests Under Estimation Risk

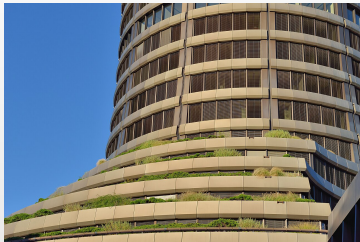
- ▣ Forecasts use  $\hat{\theta}$  estimated on  $R$  days; the test uses  $P$  out-of-sample days; the hits  $\hat{I}_t = 1\{L_t > \text{VaR}_t(\hat{\theta})\}$  are not Bernoulli( $\alpha$ ) even under a correct model
- ▣ The correction depends on the forecasting scheme (fixed, rolling, recursive) and grows with  $P/R$  (Escanciano & Olmo (2010); expansion in the Appendix)
- ▣ Fixed-scheme illustration: GARCH(1,1)-t with the S&P 500 estimates,  $R = 1000$ , 400 paths: Kupiec rejects a **correct** model in 27.5% of samples at  $P = 5000$ , against 4.0% with the true parameters
- ▣ This chapter refits on a rolling window ( $R = 1000$ , every 20 days): the fixed-scheme number shows the mechanism, not the size of our tests
- ▣ Remedy: reproduce the estimation inside the calibration, e.g. a parametric bootstrap that re-estimates the model on every simulated path (Seminar 8, B9)

## The Traffic Light: a Binomial Count Rule

| Exceptions | $P(X \leq x)$ | Zone   |
|------------|---------------|--------|
| 0          | 8.11%         | green  |
| 2          | 54.32%        | green  |
| 4          | 89.22%        | green  |
| 5          | 95.88%        | yellow |
| 7          | 99.60%        | yellow |
| 9          | 99.97%        | yellow |
| 10         | 99.99%        | red    |

- ▣ Banks using internal models backtest their daily VaR 1% against actual P&L (profit and loss) over the last 250 days (BCBS (1996))
- ▣  $X \sim \text{Binomial}(250, 0.01)$ : exceptions of a correct VaR 1% in 250 days
- ▣ Yellow starts where  $P(X \leq x) \geq 95\%$ , red where  $P(X \leq x) \geq 99.99\%$
- ▣ Only the **number** of exceptions counts, not their size or timing: simple, automatic, hard to game

## Backtesting Rules in 1996 and 2019

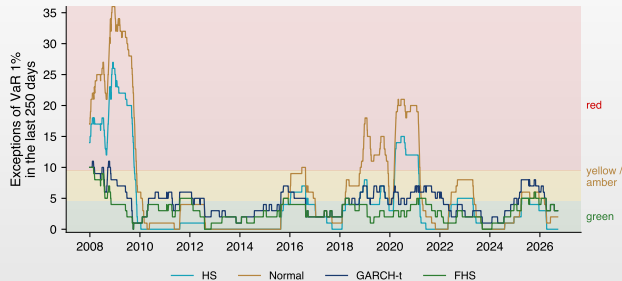


BIS tower, Basel: seat of the BCBS secretariat

Photo: Taxiarchos228 (2014); Free Art License;  
Wikimedia Commons

- **BCBS (1996):** capital on VaR 1% over 10 days
  - ▶ multiplier 3, plus 0.40–0.85 in the yellow zone and plus 1 in the red zone
- **BCBS (2019) (FRTB):** capital on ES 2.5%, backtesting still on VaR
  - ▶ bank-wide: VaR 1%, 250 days; multiplier 1.50 (green, 0–4), 1.70–1.92 (amber, 5–9), 2.00 (red, 10 or more)
  - ▶ trading desk: more than 12 exceptions of VaR 1% or 30 of VaR 2.5% in 12 months move the desk to the standardised approach
  - ▶ IMA approval also requires a P&L attribution test (not covered here)
- VaR counts stay because they are simple; ES non-elicitability (Section 5) does not prevent ES backtests (Section 4)

## The Traffic Light Through Time



- VaR 1% exceptions in the last 250 days, S&P 500; bands: green, yellow, red zones
- Share of days in red: Normal 22.3%, HS 14.7%, GARCH-t 2.0%, FHS 0.8%
- Peak: HS 27 and Normal 37 exceptions in the 250 days ending 01.12.2008; GARCH-t is yellow 46.6% of the time
- Overlapping windows keep a model penalised for a year after a crisis, when capital is already strained

## FRTB Desk Check on the Last 250 Days (Limits 12 and 30)

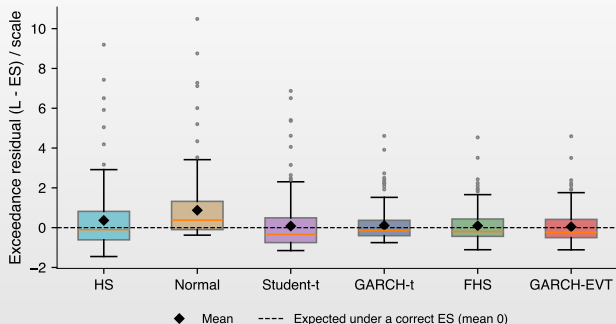
| Model     | S&P 500 | BET   | Bitcoin | EUR/RON |
|-----------|---------|-------|---------|---------|
| HS        | 0 / 2   | 4 / 7 | 3 / 4   | 1 / 5   |
| Normal    | 2 / 2   | 6 / 9 | 3 / 5   | 1 / 4   |
| Student-t | 0 / 2   | 4 / 9 | 1 / 4   | 5 / 8   |
| GARCH-t   | 3 / 6   | 4 / 9 | 3 / 5   | 4 / 5   |
| FHS       | 3 / 4   | 4 / 6 | 4 / 5   | 1 / 4   |
| GARCH-EVT | 3 / 3   | 2 / 7 | 4 / 5   | 1 / 4   |

- Cells: exceptions of VaR 1% / VaR 2.5% in the last 250 days of each series (to 18.09.2026); every model passes comfortably
- Illustration only: FRTB applies the limits to desk P&L (actual and hypothetical), not to index returns
- The last year was calm: a green light today says little about the next crisis

## McNeil and Frey (2000): Exceedance Residuals

- ES is an **average** beyond VaR: the information sits on about  $\alpha T$  tail days (124 for the S&P 500 at  $\alpha = 2.5\%$ ; 6.25 in one year, so a tail-mean s.e. of order  $1/\sqrt{\alpha T} \approx 0.40$ )
- On each breach day of VaR 2.5%:  $e_t = (L_t - \text{ES}_t)/\sigma_t$ , with  $\sigma_t > 0$  the forecast volatility, known at  $t - 1$
- If VaR **and** ES are correct:  $E[e_t \mid L_t > \text{VaR}_t, \mathcal{F}_{t-1}] = 0$ ; a wrong VaR changes the conditioning event
- $H_0$ :  $E[e_t] = 0$  against  $H_1$ :  $E[e_t] > 0$  (ES underestimated); one-sided  $t$ -statistic
- Few, fat-tailed residuals:  $p$ -value from an i.i.d. bootstrap of the centred residuals (McNeil & Frey (2000)), which assumes independent residuals; otherwise a block bootstrap

## Exceedance Residuals on the S&P 500



- Distribution of  $(L_t - ES_t) / \sigma_t$  on breach days of VaR 2.5%; diamonds: means
- Means: Normal 0.875, HS 0.363, GARCH-t 0.106, FHS 0.093, GARCH-EVT 0.045
- Bootstrap  $p$ : Normal  $<0.001$ , HS  $<0.001$ , GARCH-t 0.023, FHS 0.096, GARCH-EVT 0.246
- The Normal distribution underestimates the tail mean by almost one volatility unit; the outliers (up to 10 units) are the days that decide capital adequacy

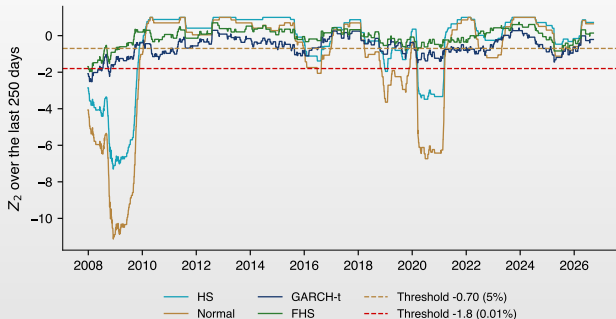
## Acerbi and Szekely (2014): Testing ES Directly

- ▣ Breach indicator at the ES level:  $I_t = 1\{L_t > \text{VaR}_t^\alpha\}$ ,  $N_T = \sum_t I_t$ ,  $\alpha = 0.025$
- ▣  $Z_1 = 1 - \frac{1}{N_T} \sum_t \frac{L_t I_t}{\text{ES}_t}$ : is the average tail loss as large as forecast? (conditional on breaches)
- ▣  $Z_2 = 1 - \frac{1}{T_\alpha} \sum_t \frac{L_t I_t}{\text{ES}_t}$ : tests frequency and size together
- ▣ Under  $H_0$  (correct forecast distribution)  $E[Z_2] = 0$ ; underestimated risk gives  $Z_2 < 0$  (Acerbi & Szekely (2014))
- ▣ For  $T = 250$  and many distributions, the 5% and 0.01% critical values are close to  $-0.70$  and  $-1.8$ : reference thresholds for an ES traffic light

## Getting $p$ -Values by Simulation

- The null distribution of  $Z_2$  depends on the forecast distributions  $F_t$ , which the model supplies
- Algorithm (repeat  $M = 2000$  times)
  - ▶ For each day draw  $U_t \sim \text{Uniform}(0,1)$  and set  $L_t^* = F_t^{-1}(U_t)$
  - ▶ Compute  $Z_2^*$  from the simulated losses with the same VaR and ES forecasts
- $p$ -value = share of simulated  $Z_2^*$  at or below the observed  $Z_2$
- Only tail draws matter: for HS and FHS we store the 25 largest window losses of each day
- S&P 500, full sample:  $Z_2 = -0.618$  for GARCH-t ( $p < 0.001$ ) versus  $-0.091$  for FHS ( $p = 0.172$ ); critical value about  $-0.152$

## $Z_2$ Over Rolling 250-Day Windows



- $Z_2$  of the ES 2.5% forecasts over the last 250 days, S&P 500; the illustrative reference thresholds  $-0.70$  and  $-1.8$
- Minimum: Normal  $-11.13$ , HS  $-7.30$  (window ending 01.12.2008), GARCH-t  $-2.52$ , FHS  $-1.97$
- Windows below  $-1.8$ : Normal 21.9%, HS 15.8%, GARCH-t 3.0%, FHS 0.7%; GARCH-t below  $-0.70$  in 39.6%
- Checking this every day is repeated testing: false alarms accumulate (sequential monitoring, Appendix)

## Du and Escanciano (2017): Cumulative Violations

- ▣ PIT:  $u_t = F_t(L_t)$ , uniform on  $(0, 1)$  if the forecast distribution is correct
- ▣ Cumulative violation:  $H_t = \frac{1}{\alpha} (u_t - (1 - \alpha)) 1\{u_t > 1 - \alpha\}$ : how deep into the tail the loss went
- ▣ Under  $H_0$ :  $E[H_t] = \alpha/2$  and  $\text{Var}(H_t) = \alpha(1/3 - \alpha/4)$
- ▣ Unconditional test:  $U = \sqrt{T} (\bar{H} - \alpha/2) / \sqrt{\alpha(1/3 - \alpha/4)} \xrightarrow{d} N(0, 1)$
- ▣ Conditional test: Box–Pierce statistic on the first 5 autocorrelations of  $H_t$ ,  $\chi^2_5$  (Du & Escanciano (2017))

## All ES Tests at a Glance

|           | McNeil-Frey |        |         |         | Acerbi-Szekely $Z_2$ |        |         |         | Du-Escanciano $U$ |        |         |         |
|-----------|-------------|--------|---------|---------|----------------------|--------|---------|---------|-------------------|--------|---------|---------|
| HS        | <0.001      | 0.48   | 0.70    | 0.46    | <0.001               | 0.13   | 0.86    | 1.00    | <0.001            | 0.36   | 0.14    | <0.001  |
| Normal    | <0.001      | <0.001 | <0.001  | <0.001  | <0.001               | <0.001 | 0.001   | 0.97    | <0.001            | <0.001 | <0.001  | 0.25    |
| Student-t | 0.17        | 0.01   | 1.00    | 0.47    | 0.002                | 0.003  | 0.62    | 0.92    | <0.001            | <0.001 | 0.93    | 0.05    |
| GARCH-t   | 0.02        | 0.04   | 0.50    | 0.41    | <0.001               | <0.001 | 0.05    | 0.63    | <0.001            | <0.001 | 0.13    | 0.86    |
| FHS       | 0.10        | 0.63   | 0.60    | 0.75    | 0.17                 | 0.19   | 0.35    | 0.97    | 0.09              | 0.46   | 0.95    | 0.09    |
| GARCH-EVT | 0.25        | 0.57   | 0.29    | 0.69    | 0.09                 | 0.21   | 0.58    | 0.98    | 0.24              | 0.54   | 0.70    | 0.11    |
|           | S&P 500     | BET    | Bitcoin | EUR/RON | S&P 500              | BET    | Bitcoin | EUR/RON | S&P 500           | BET    | Bitcoin | EUR/RON |

- $p$ -values for ES 2.5%: McNeil–Frey (bootstrap),  $Z_2$  (simulation), Du–Escanciano  $U$ ; the Normal distribution is rejected by all three on S&P 500, BET and Bitcoin; FHS and GARCH-EVT pass all three everywhere
- Different nulls and inputs, different decisions: Student-t on the S&P 500 has McNeil–Frey  $p = 0.173$  but  $Z_2$   $p = 0.002$ ; these are calibration tests, not rankings
- GARCH-t fails  $Z_2$  on equities ( $p < 0.001$ ,  $< 0.001$ ) while FHS passes; EUR/RON: most models over-predict ES ( $Z_2 > 0$ )

## Multinomial VaR Backtests: an Implicit ES Test

- ES 2.5% averages the quantiles beyond VaR 2.5%: test several VaR levels at once (Kratz, Lok & McNeil (2018))
- $N = 4$  tail levels  $\alpha_j = 2.5\% (1 - (j - 1)/4)$ : 2.5%, 1.875%, 1.25%, 0.625%; each day falls in one of 5 cells via the PIT  $u_t$
- Pearson statistic  $S = \sum_{j=0}^4 (O_j - Tp_j)^2 / (Tp_j) \xrightarrow{d} \chi_4^2$ , cell probabilities  $p_0 = 97.5\%$ ,  $p_1 = \dots = p_4 = 0.625\%$

| $p$ -value | S&P 500 | BET    | Bitcoin | EUR/RON |
|------------|---------|--------|---------|---------|
| HS         | <0.001  | 0.725  | 0.564   | 0.004   |
| Normal     | <0.001  | <0.001 | <0.001  | <0.001  |
| Student-t  | <0.001  | 0.024  | 0.015   | 0.172   |
| GARCH-t    | <0.001  | <0.001 | 0.227   | 0.568   |
| FHS        | 0.008   | 0.846  | 0.413   | 0.392   |
| GARCH-EVT  | 0.287   | 0.864  | 0.969   | 0.240   |

- Only GARCH-EVT passes in all four markets; S&P 500 GARCH-t cells (4767, 36, 46, 50, 60) against 31 expected in each tail cell: too many days in every tail cell, most in the deepest

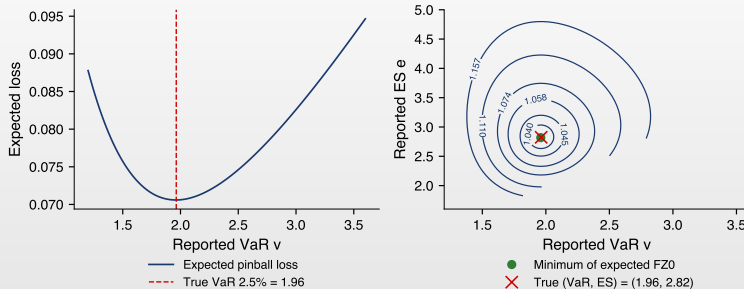
## Testing Is Not Ranking: Consistent Scores

- A backtest asks whether a model is **acceptable** (several can pass, all can fail); comparative backtesting asks which is **better** (Nolde & Ziegel (2017))
- Ranking needs a **scoring function**  $S(x, L)$ , the loss of reporting  $x$  when  $L$  occurs; compare  $\bar{S} = \frac{1}{T} \sum_t S(x_t, L_t)$ , lower is better
- $S$  is **consistent** for  $\theta(F)$ , and  $\theta$  is **elicitable**, if  $\theta(F) = \arg \min_x E_F[S(x, L)]$  (Gneiting (2011)): the truth minimises the expected score
- VaR: pinball loss  $S(v, L) = (1\{L > v\} - \alpha)(L - v)$ ; for  $E|L| < \infty$  and  $f > 0$  at the quantile
  - ▶  $\frac{\partial}{\partial v} E_F[S(v, L)] = F(v) - (1 - \alpha) = 0 \Rightarrow v = F^{-1}(1 - \alpha) = \text{VaR}_\alpha$ , and  $\partial^2 / \partial v^2 = f(v) > 0$
- ES is **not** elicitable on its own: no scoring function of  $e$  alone is minimised by the true ES (Gneiting (2011))

## VaR and ES Together: Fissler and Ziegel (2016)

- The pair (VaR, ES) is jointly elicitable: [Fissler & Ziegel \(2016\)](#) characterise the whole family of consistent scoring functions
- FZ0 loss ([Patton, Ziegel & Chen \(2019\)](#)), with losses positive,  $0 < v \leq e$ ,  $E|L| < \infty$  and a true ES above 0
  - ▶ 
$$\text{FZ0}(v, e; L) = \frac{1\{L > v\}(L - v)}{\alpha e} + \frac{v}{e} + \ln e - 1$$
- First term: penalty for breaches, scaled by ES; last terms: penalty for reporting a large ES
- Differences of FZ0 losses do not depend on the units of the losses, so they can be compared across assets
- Lower average FZ0 = better joint VaR and ES forecast

## Elicitability in Pictures

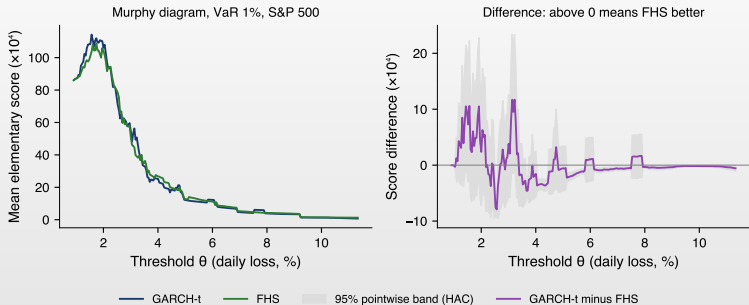


- Student-t losses with  $\nu = 4$  and unit variance,  $\alpha = 2.5\%$ ; expected losses computed on 2 million draws
- Left: the expected pinball loss is lowest at the true VaR 2.5%, 1.96; right: the expected FZ0 loss is lowest at (1.96, 2.82), the true (VaR, ES) up to the grid
- A forecaster who knows the truth has no incentive to report anything else: this is what makes ranking by average score meaningful

## Is the Ranking Robust to the Score? Elementary Scores

- For a quantile there are infinitely many consistent scores:  $S(v, L) = (1\{L \leq v\} - \tau)(g(v) - g(L))$ , any increasing  $g$ ,  $\tau = 1 - \alpha$
- Each is a mixture of **elementary scores** (Ehm, Gneiting, Jordan & Krüger (2016))
  - ▶  $S_\theta(v, L) = (1\{L < v\} - \tau)(1\{\theta < v\} - 1\{\theta < L\})$ ,  $\theta \in \mathbb{R}$ ; the pinball loss is  $\int S_\theta d\theta$
- Forecast  $A$  beats  $B$  under **every** consistent score iff  $\bar{S}_\theta(A) \leq \bar{S}_\theta(B)$  for all  $\theta$
- **Murphy diagram**: plot  $\bar{S}_\theta$  against  $\theta$ ; crossing curves mean the ranking depends on the chosen score

## Murphy Diagram: GARCH-t versus FHS

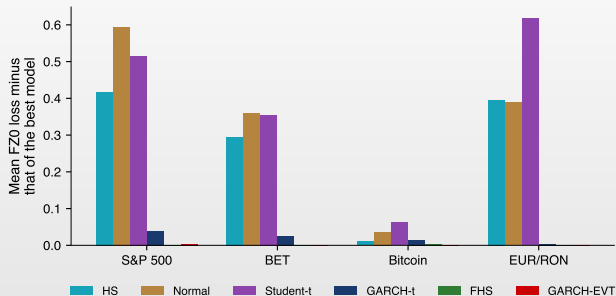


- S&P 500, VaR 1%, 2007–2026;  $\theta$  from 0.9% to 11.3%; pinball ( $\times 10^4$ ): GARCH-t 3.849, FHS 3.849, a tie under the usual score
- Below  $\theta = 5\%$  the curves cross: FHS better on 50% of thresholds, band containing zero on 82%; above 5% GARCH-t lower on 89%: no dominance
- The bands are pointwise, not a simultaneous test of dominance: state the score before looking at the results

## The Diebold–Mariano Test on FZ0 Losses

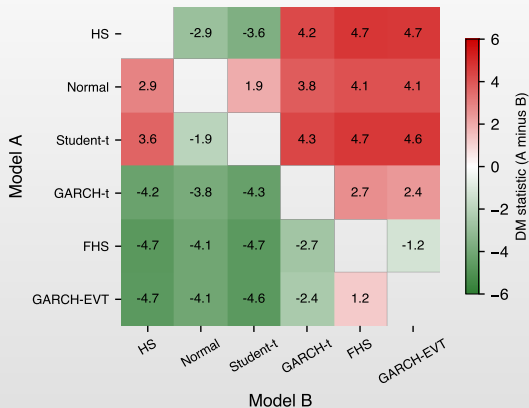
- Loss differential of models  $A$  and  $B$ :  $d_t = \text{FZ0}_{A,t} - \text{FZ0}_{B,t}$
- $H_0$ :  $E[d_t] = 0$  (equal predictive accuracy), Diebold & Mariano (1995)
- $$\text{DM} = \frac{\bar{d}}{\sqrt{\widehat{\text{LRV}}(d_t)/T}} \xrightarrow{d} N(0, 1)$$
- $\widehat{\text{LRV}}$ : long-run variance with Newey–West weights (Newey & West (1987)), 9 lags; tail losses cluster, so  $d_t$  is autocorrelated
- $\text{DM} > 0$ : model  $A$  has the larger average loss (worse);  $|\text{DM}| > 1.96$ : significant at 5%

## Average FZ0 Loss by Model



- Mean FZ0 loss of the (VaR 2.5%, ES 2.5%) forecasts minus that of the best model in each market
- Lowest: FHS on the S&P 500; GARCH-EVT on BET, Bitcoin and EUR/RON
- Unconditional models lose 0.3–0.6 FZ0 units on S&P 500, BET and EUR/RON; on Bitcoin the largest gap is 0.062
- GARCH-t trails FHS on the S&P 500 by 0.039: the size of a gap alone says nothing about whether the data can separate the models; test it

## Pairwise Diebold–Mariano Tests, S&P 500



- DM of model  $A$  (row) minus  $B$  (column) on FZ0 losses; red:  $A$  worse, green:  $A$  better
- Against FHS: HS DM = 4.70 and GARCH-t DM = 2.65, both significantly worse; GARCH-EVT DM = 1.24, indistinguishable
- Normal minus HS: DM = 2.88: a slow empirical quantile still beats a thin tail
- Fifteen pairwise tests raise a multiple-testing problem: which models survive **jointly**? (MCS, below)

## Comparing Forecasting Methods: Giacomini–White

- DM targets the population accuracy of **models**; with rolling 1000-day windows and re-estimated parameters we compare **forecasting methods**, and estimation error never vanishes
- **Giacomini & White (2006)**:  $H_0: E[d_t | \mathcal{F}_{t-1}] = 0$  for the score differential  $d_t = \text{FZ0}_{A,t} - \text{FZ0}_{B,t}$ ; valid for nested models and finite windows
- Test function  $h_{t-1} = (1, d_{t-1})'$ ,  $Z_t = h_{t-1} d_t$ :  $\text{GW} = T \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \xrightarrow{d} \chi_2^2$ ,  $\hat{\Omega} = T^{-1} \sum_t Z_t Z_t'$  (one step ahead);  $h_{t-1} = 1$  gives the DM form
- S&P 500 against FHS: GARCH-EVT  $\text{GW} = 7.38$  ( $p = 0.025$ ), GARCH-t 7.59 ( $p = 0.022$ ), HS 53.38 ( $p < 0.001$ )
- GARCH-EVT, a tie under DM, is rejected conditionally: equal conditional predictive ability fails for these instruments; which moment drives it needs a separate test

## Three-Zone Comparative Backtesting (Nolde and Ziegel)

- Internal model  $A$  against a standard model  $S$  (here HS), mean FZ0 loss; two one-sided tests at 5% (Nolde & Ziegel (2017))
- $H_0^-$ :  $A$  at least as good as  $S$ ;  $H_0^+$ :  $A$  at most as good. **Green**: reject  $H_0^+$ ; **red**: reject  $H_0^-$ ; **yellow**: neither
- The burden of proof is reversed: the bank must show that its model is better, not the supervisor that it is worse

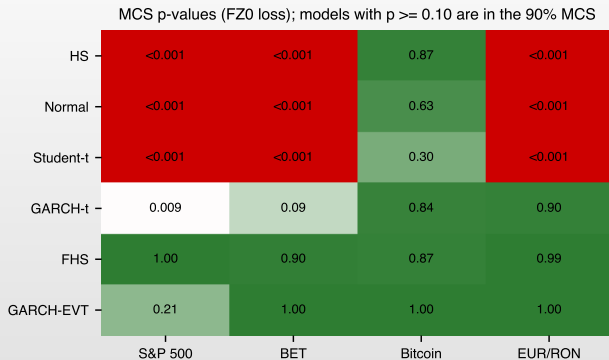
| DM (A minus HS) | S&P 500 | BET   | Bitcoin | EUR/RON |
|-----------------|---------|-------|---------|---------|
| GARCH-t         | −4.23   | −5.18 | 0.09    | −6.66   |
| FHS             | −4.70   | −5.49 | −0.17   | −6.94   |
| GARCH-EVT       | −4.68   | −5.53 | −0.25   | −6.92   |

- Colour = zone. Green everywhere except Bitcoin, where every model is yellow: with its noisy tail, no model can prove it beats HS

## The Model Confidence Set

- ▣ Hansen, Lunde & Nason (2011): a set of models built to contain the best models with asymptotic probability at least  $1 - \alpha_{MCS}$ , under the method's regularity conditions
- ▣ Algorithm
  - ▶ Test  $H_0$ : all models in the current set have equal expected loss
  - ▶ If rejected, eliminate the model with the largest standardised average loss and repeat
  - ▶ Critical values from a block bootstrap (blocks of 10 days, 1000 replications)
- ▣ MCS  $p$ -value of a model: the largest test  $p$ -value up to its elimination; keep models with  $p \geq 0.10$
- ▣ A large MCS is an honest answer: the data do not separate the models

## Which Models Survive?



- MCS  $p$ -values for the FZ0 loss of (VaR 2.5%, ES 2.5%), per market; models with  $p \geq 0.10$  form the 90% MCS
- S&P 500: FHS and GARCH-EVT (MCS  $p$  0.21); GARCH-t out ( $p$  0.01)
- BET: FHS and GARCH-EVT, GARCH-t just out ( $p$  0.09); EUR/RON: the three conditional models
- Bitcoin: all six models stay: the data cannot separate them; non-rejection is not proof of equal performance

## Split Conformal Quantiles: a Distribution-Free Promise

- All models so far assume a distribution (Normal, t, GPD) or a stable history (HS); **conformal prediction** turns any forecast into a quantile with a coverage guarantee (Lei et al. (2018), Angelopoulos & Bates (2021))
- Score  $s_t = L_t / \hat{\sigma}_t$ ,  $\hat{\sigma}_t$  the standard deviation of the previous 250 returns; calibration set: the last  $n = 250$  scores, sorted  $s_{(1)} \leq \dots \leq s_{(n)}$
- Conformal VaR:  $\text{VaR}_t = \hat{\sigma}_t s_{(k)}$  with  $k = \lceil (n+1)(1-\alpha) \rceil$ ;  $n = 250$ ,  $\alpha = 0.01$ :  $k = 249$ , the second largest score
- Guarantee if the  $n+1$  scores are **exchangeable** (joint law invariant to reordering):  
 $1 - \alpha \leq P(L_t \leq \text{VaR}_t) \leq 1 - \alpha + \frac{1}{n+1}$  (no ties)
- Volatility clustering breaks exchangeability, and here  $\hat{\sigma}_t$  comes from overlapping rolling windows: the guarantee is approximate, unlike genuine split conformal with a score fitted once on a separate sample (extensions: Appendix)

## Adaptive Conformal Inference

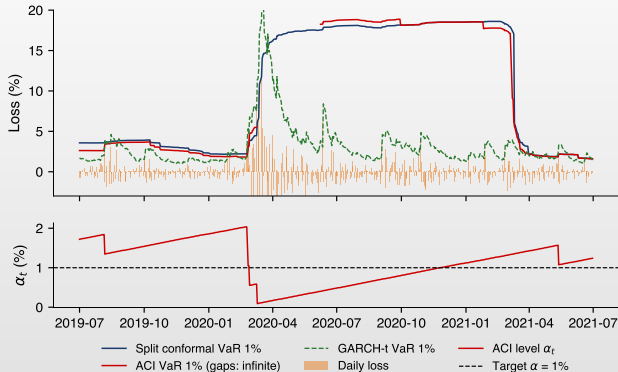


Emmanuel Candès, co-author of adaptive conformal inference

Photo: CmicHEL67 (2026); CC BY-SA 4.0;  
Wikimedia Commons

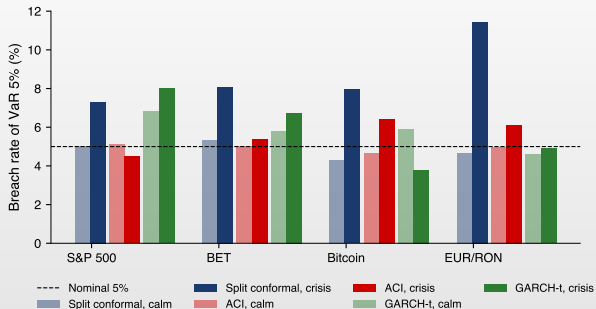
- Gibbs & Candès (2021): update the working level after each day,  $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ ,  $\text{err}_t = 1\{L_t > \text{VaR}_t\}$
- After a breach  $\alpha_t$  falls (VaR rises); after calm days it slowly rises again; we use  $\gamma = 0.005$
- Guarantee for any sequence:  $\left| \frac{1}{T} \sum_t \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$ , a long-run **average**, not each day or each crisis
- If  $\alpha_t < 1/(n+1)$ , the required quantile does not exist: the VaR becomes infinite

## Conformal VaR Through COVID-19



- S&P 500, July 2019–June 2021: split conformal, ACI and GARCH-t VaR 1% (top), ACI level  $\alpha_t$  (bottom)
- 20.02–30.04.2020, 50 days, 0.5 breaches expected: split conformal 6, ACI 4 (8% of days)
- ACI set  $\text{VaR} = \infty$  on 37 of these days: no further breaches, but an infinite VaR carries no information
- With March 2020 in its calibration set, the split conformal VaR stays near 18% for a year; GARCH-t adapts within days

## Coverage in Calm and Crisis Periods



- VaR 5%; crisis day: previous 20-day volatility above the 90th percentile of its values over the previous 1000 days (known in advance)
- Split conformal, all days: S&P 500 5.34%, BET 5.63%, Bitcoin 4.58%, EUR/RON 5.06%; crisis days: 7.29%, 8.08%, 7.95%, 11.43%
- ACI brings equity crisis rates near 5% (4.49%, 5.38%); for VaR 1% it keeps the average only by an infinite VaR on 13.4% of S&P 500 days: marginal coverage is not crisis coverage

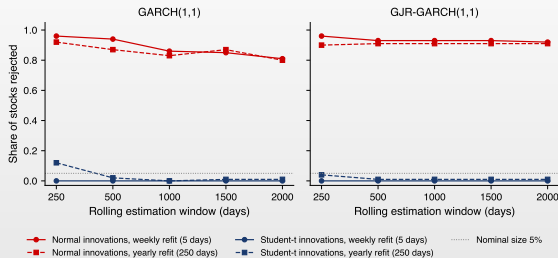
## Can ES Be Backtested Using Only ES Forecasts?

- **Bayer & Dimitriadis (2022)**: can ES forecasts be backtested with nothing but the realised returns and the ES forecasts themselves?
- **Why it matters**: FRTB puts capital on ES 2.5% (Section 3)
  - ▶ The ES tests of Section 4 also need VaR forecasts, the volatility or the whole forecast distribution; supervisors do not necessarily receive these inputs, and a rejection of such a joint test does not say which input is wrong (Section 1.1, Definition 1.1 of the paper)
- **Idea**: a Mincer–Zarnowitz regression of the returns on the ES forecasts,  $H_0$ : intercept 0, slope 1
  - ▶ ES alone is not elicitable (Section 5), but (VaR, ES) is jointly elicitable (**Fissler & Ziegel (2016)**), so a joint quantile–ES regression can be estimated (**Patton, Ziegel & Chen (2019)**)
- **Returns convention**:  $\hat{e}_t = -\widehat{ES}_t < 0$ ,  $\tau = 2.5\%$ ; ES equation  $r_t = \gamma_1 + \gamma_2 \hat{e}_t + u_t^e$ ,  $ES_\tau(u_t^e | \mathcal{F}_{t-1}) = 0$  (Eq. 1.4–1.6)
  - ▶ **Auxiliary ESR**: quantile equation on  $(1, \hat{v}_t)$ , needs VaR forecasts; **Strict ESR**: quantile equation on  $(1, \hat{e}_t)$ , ES forecasts only
  - ▶ **Intercept ESR** (Eq. 1.14–1.15):  $r_t - \hat{e}_t = \gamma_1 + u_t^e$ ; one-sided  $t$ -test of  $H_0: \gamma_1 \geq 0$  against  $\gamma_1 < 0$  (ES underestimated)

## Estimation, Covariance and Monte Carlo Evidence

- M-estimator minimising the FZ0-type loss of Section 5 (Eq. 1.16–1.18); asymptotics under misspecification (Theorem 1.4); R packages [esback](#) and [esreg](#)
- Two assumptions to keep in mind
  - ▶ Strict and Intercept ESR: the quantile equation is correct only if VaR is an affine function of ES (e.g. pure scale models) (Section 1.2)
  - ▶ The misspecification-robust sandwich covariance assumes a location-scale model for the conditional distribution (Section 1.5)
- Design (Section 2): 10 000 replications,  $T$  from 250 to 5 000; EGARCH-t calibrated to the S&P 500, AR(1)-GARCH(1,1), two Student-t GAS models; competitors [McNeil & Frey \(2000\)](#) and the calibration tests of [Nolde & Ziegel \(2017\)](#)
- Size at 5%, EGARCH-t: Strict ESR 0.09 ( $T = 250$ ) and 0.05 ( $T = 1\,000$ ) with the robust covariance, 0.24 and 0.11 with the classical one; the simple conditional calibration test 0.29 and 0.14 (Table 1)
- Power: exceedance-residual tests barely react to wrong volatility dynamics or to forecasts for a wrong level  $\tilde{\tau} \neq \tau$ , which ESR tests detect (Fig. 3): a bank could report ES for a milder level and pass the older tests

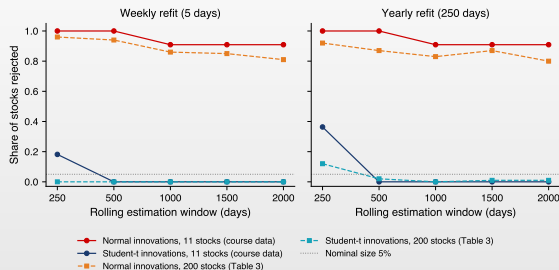
## Table 3 of the Paper: Tails Matter More Than Refits



- Share of the 200 largest S&P 500 stocks with ES 2.5% rejected by the one-sided Intercept ESR at 5%, 2010–2019; windows 250–2 000 days, refits every 5 (solid) or 250 days (dashed); [Bayer & Dimitriadis \(2022\)](#), Table 3
- Normal innovations: 0.80–0.96 rejected in all 50 window–refit cells; Student-t: up to 0.12 with 250-day windows, at most 0.01 with 1 000 days or more
- For these stocks and this period: fat tails and a long window matter; the refit frequency matters only with short windows

 MFM\_ch8\_esr\_case

## Table 3 Replicated on the Course Data



- Solid: 11 large S&P 500 stocks with prices from 2002 (Apple, Microsoft, Amazon, NVIDIA, Cisco, JPMorgan Chase, Bank of America, Citigroup, Wells Fargo, Goldman Sachs, Morgan Stanley), January 2010 – August 2019 (2432 days); dashed: Bayer & Dimitriadis (2022), Table 3
- GARCH(1,1) with a constant mean (the paper does not state the mean equation), the 25 window–refit cells, one-sided Intercept ESR at 5% with [esback](#); one stock moves a share by 0.09
- Normal: 0.82–1.00 rejected (paper 0.80–0.96); Student-t: at most 0.00 from 1000 days (paper 0.01), but 0.09–0.55 at 250 days (paper 0.00–0.12): the central result replicates, the short-window shares are higher

## An Open Question: Crisis-Conditional Coverage

- Distribution-free VaR guarantees are **marginal**: split conformal needs exchangeability, ACI controls only the long-run average (Section 7)
- Formal question, with a crisis flag  $C_t \in \mathcal{F}_{t-1}$  known ex ante
  - ▶  $H_0: P(L_t > \text{VaR}_t \mid C_t = 1) = \alpha$  against  $H_1: P(L_t > \text{VaR}_t \mid C_t = 1) \neq \alpha$
  - ▶ Falsified by a crisis breach rate outside the Binomial acceptance region, after correcting for dependent breaches and for multiplicity
- One pass of the discovery loop with an AI assistant (an LLM such as Claude, ChatGPT or Gemini)
  - ▶ literature with DOIs checked on Crossref; three rival explanations (stale scale, changing tail shape, regime-selection artefact); replicate a known number first; pre-register the tests; ask the assistant for a hostile referee report
  - ▶ Reusable prompts and a research checklist: Appendix
- Stakes: capital is needed exactly on the days when marginal guarantees say least

## Mini-Case: Does a GARCH Scale Repair Crisis Coverage?

- Assistant proposal: score  $s_t = (L_t + \mu_t)/\sigma_t$  with the GARCH-t forecasts;  $H_0$ : crisis breach rate of VaR 5% = 5%, exact Binomial test, Holm correction across the four markets

| Market  | Crisis days | Split conformal | Holm $p$ | GARCH-scaled | Holm $p$ |
|---------|-------------|-----------------|----------|--------------|----------|
| S&P 500 | 613         | 7.01%           | 0.052    | 2.77%        | 0.037    |
| BET     | 506         | 8.10%           | 0.009    | 3.95%        | 0.309    |
| Bitcoin | 150         | 8.00%           | 0.092    | 1.33%        | 0.111    |
| EUR/RON | 245         | 11.43%          | <0.001   | 2.45%        | 0.153    |

- Sample change first: the GARCH-scaled scores need 250 calibration days, so the common S&P 500 sample has 4 709 days and 613 crisis days (Section 7: 713), and the split conformal crisis rate becomes 7.01% (Section 7: 7.29%)
- The GARCH scale pushes every crisis rate below 5%; Holm rejects only on the S&P 500 (2.77%); elsewhere this is non-rejection, not proof of coverage: Binomial  $p$ -values ignore clustered breaches and the crisis samples are small

 ai\_discovery\_case.py

## Which Model Would You Provisionally Select?



National Bank of Romania, Bucharest

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- ▣ **Calibration:** only FHS and GARCH-EVT pass  $Z_2$  at 5% and stay in the 90% MCS in all four markets (scorecard in the Appendix)
- ▣ **Comparison:** FHS has the lowest FZ0 on the S&P 500, GARCH-EVT elsewhere; the two are not significantly different, and on Bitcoin no model can be separated from HS
- ▣ **Unresolved:** DQ rejects every S&P 500 model, crisis-conditional coverage fails, and one year of data has little power
- ▣ **Provisional choice:** a GARCH filter with a flexible tail, with the other as challenger, re-examined as evidence accrues; a Romanian book (BET and EUR/RON): Seminar 8, C1

## Check Yourself, and Next: Chapter 9

### Retrieval questions

- Why does the rule " $LR_{uc} > 3.84$ " reject a correct VaR 1% in about one year in ten?
- Why can a model with the right number of breaches still fail?
- Why can ES be backtested but not ranked on its own?
- What does ACI guarantee, and what not?

### Chapter 9: realised volatility

- Measure daily variance from intraday returns instead of forecasting it from daily data only
- Evaluate volatility forecasts with robust scores (QLIKE) and the MCS of this chapter

## Appendix: Further Reading and Resources

- ▣ Fissler & Ziegel (2016): VaR and ES are jointly elicitable, the basis of the FZ0 loss; Nolde & Ziegel (2017): comparative backtests and the traffic light for VaR and ES; Acerbi & Szekely (2014): model-free ES backtests, including  $Z_2$
- ▣ Entropy-based VaR and ES forecasts: Pele, Lazar & Dufour (2017), Pele & Mazurencu-Marinescu-Pele (2019), Pele, Lazar & Mazurencu-Marinescu-Pele (2019)
- ▣ When two ES forecasts can be told apart: Pele & Mazurencu-Marinescu-Pele (2026) find that roughly one in five pairwise ES comparisons in global markets is precision-fragile, a consequence of the small effective tail sample  $\alpha T$
- ▣ Textbooks: Franke, Härdle & Hafner (2019) (VaR and backtesting chapters) and Härdle, Chen & Overbeck (2017)
- ▣ Quantinar: Machine learning in Financial Risk
- ▣ Code and data: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_08](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_08); S&P 500, BET, Bitcoin from `data/market`; EUR/RON: BNR reference rate

## Appendix: Estimation Risk, the Fixed-Scheme Expansion

- Forecasts use  $\hat{\theta}$  estimated on  $R$  days; the test uses  $P$  out-of-sample days
- Fixed scheme ([Escanciano & Olmo \(2010\)](#)):  $\frac{1}{\sqrt{P}} \sum_t (\hat{I}_t - \alpha) = \frac{1}{\sqrt{P}} \sum_t (I_t - \alpha) + \sqrt{P/R} D' \sqrt{R} (\hat{\theta} - \theta_0) + o_p(1)$ 
  - ▶  $D = E [\partial P(L_t > \text{VaR}_t(\theta) \mid \mathcal{F}_{t-1}) / \partial \theta]$  at  $\theta_0$ ; the second term vanishes only if  $P/R \rightarrow 0$
  - ▶ The estimation term adds a variance of order  $P/R$  to  $\sqrt{P}(\hat{\pi} - \alpha)$ , so  $\chi^2$  critical values are too small when  $P/R$  is large
- Rolling and recursive schemes change the form of the correction; the expansion assumes smooth  $\text{VaR}_t(\theta)$ , a  $\sqrt{R}$ -consistent  $\hat{\theta}$  and stationarity
- Remedies: the corrected variance or subsampling of [Escanciano & Olmo \(2010\)](#), or a parametric bootstrap that re-estimates the model on every simulated path
- Fixed scheme, 400 paths,  $R = 1000$ : Kupiec rejection 11.5% / 12.8% / 27.5% at  $P = 250 / 1000 / 5000$  with estimated parameters, 10.0% / 5.2% / 4.0% with the true ones; Monte Carlo s.e. about 1.1 points at 5%

## Appendix: Osband's Principle and Identification Functions

- ▣ Osband's principle (Fissler & Ziegel (2016)): consistent scores are built from an identification function  $V$ , with  $E_F[V(x, L)] = 0$  only at the true value
- ▣ VaR:  $V(v, L) = 1\{L \leq v\} - (1 - \alpha)$ ; the pinball loss is its integral in  $v$
- ▣ (VaR, ES):  $V_1 = \alpha - 1\{L > v\}$  and  $V_2 = e - v - \frac{1}{\alpha}1\{L > v\}(L - v)$ ; their expectations vanish only at the true pair
- ▣ The level sets of ES are not convex under mixtures, so no identification function of  $e$  alone exists (Seminar 8, A9)
- ▣ FZ0 is the member of the FZ family whose differences are invariant to the scale of the losses (Patton, Ziegel & Chen (2019))

## Appendix: Sequential Monitoring of Risk Forecasts

- A regulator checks the forecasts every day: repeating a fixed-sample test (e.g. rolling  $Z_2$ ) inflates the probability of at least one false alarm
- Hoga & Demetrescu (2023): monitoring statistics for VaR and ES with boundary functions that keep the probability of any false alarm over the whole horizon below the nominal level
- Compared with the traffic light: evidence accumulates from the start of monitoring instead of a single 250-day snapshot
- The ES regression backtest (Bayer & Dimitriadis (2022)) of the case study can be monitored the same way; both address the one-shot design of the traffic light

## Appendix: Conformal Prediction Beyond Exchangeability

- Barber, Candès, Ramdas & Tibshirani (2023): without exchangeability, the coverage gap is bounded by weighted total-variation distances between the score vector and the same vector with the test score swapped with calibration score  $i$
- How large these distances are under volatility clustering is an empirical question
- Gibbs, Cherian & Candès (2025): exact coverage conditional on a pre-specified class of covariate shifts is attainable; which class captures a financial crisis is open
- Choosing the ACI step  $\gamma$ : Zaffran et al. (2022) aggregate ACI experts over a grid of  $\gamma$  values

## Appendix: Case Study, Complete Key Results

| Finding                                                                                                                                                                                            | Source  | Reading                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------|
| Size at nominal 5%, EGARCH-t: Strict ESR 0.09 ( $T = 250$ ) and 0.05 ( $T = 1000$ ) with the robust covariance; 0.24 and 0.11 with the classical one                                               | Table 1 | The robust covariance improves size control; at $T = 250$ the size is still 0.09                                     |
| Strict and Auxiliary ESR sizes differ by at most 0.01 for every process and sample size, including AR(1)-GARCH with $\phi = 0.5$                                                                   | Table 1 | In these designs, dropping the VaR forecasts barely changes the size; power is compared separately (Fig. 2–3)        |
| Simple conditional calibration test: size 0.29 at $T = 250$ and 0.14 at $T = 1000$ (EGARCH-t)                                                                                                      | Table 1 | A competitor with high raw power can owe it to over-rejection                                                        |
| Exceedance-residual tests barely react to wrong volatility dynamics or to forecasts for a wrong level $\tilde{\tau} \neq \tau$ ; ESR tests detect every misspecification                           | Fig. 3  | A bank could report ES for a milder level and pass the older tests                                                   |
| Normal innovations: rejected for 80%–96% of the stocks in every window and refit design                                                                                                            | Table 3 | Normal tails understate ES for most stocks                                                                           |
| Student-t: GARCH 0.12 with a 250-day window and yearly refit; at most 0.01 with 1000 days or more, for GARCH and GJR-GARCH; leverage: from 500 days GJR-GARCH-t and GARCH-t differ by at most 0.01 | Table 3 | For these 200 stocks and 2010–2019: fat tails and windows of 1000 days or more give rejection shares of at most 0.01 |

## Appendix: Case Study, Limitations and Extensions

- Limits stated by the authors
  - ▶ Strict and Intercept ESR: the quantile equation is misspecified unless VaR is an affine function of ES (Section 1.2); Auxiliary and Strict ESR are two-sided only (Section 1.3)
  - ▶ The classical covariance over-rejects in small samples (Section 2.1)
- Replication with the course data (Seminar 8, C3)
  - ▶ Run the Strict ESR (two-sided) and the Intercept ESR (one-sided) on the ES 2.5% forecasts of the six models in the four markets; compare with  $Z_2$  and McNeil–Frey
  - ▶ Extend the Table 3 replication: GJR-GARCH, more stocks and the period after August 2019
- Open: a one-year supervisory window
  - ▶ At  $T = 250$  the two-sided Intercept ESR has size 0.04 with the robust covariance and 0.15 with the classical one (Table 1, EGARCH-t); the one-sided size is reported only for  $T = 2500$  (0.02, Table 2)
  - ▶ What is the one-sided size at one year, and can a bootstrap of the  $t$ -statistic hold it at 5% without losing the power of Fig. 4?

## Appendix: Scorecard Across Four Markets

- Each cell: breach rate of VaR 1% in % /  $Z_2$   $p$ -value / MCS  $p$ -value (FZ0)

| Model     | S&P 500                | BET                    | Bitcoin             | EUR/RON               |
|-----------|------------------------|------------------------|---------------------|-----------------------|
| HS        | 1.67 / <0.001 / <0.001 | 1.17 / 0.134 / <0.001  | 0.77 / 0.863 / 0.87 | 0.51 / 1.000 / <0.001 |
| Normal    | 2.70 / <0.001 / <0.001 | 1.90 / <0.001 / <0.001 | 1.74 / 0.001 / 0.63 | 1.06 / 0.973 / <0.001 |
| Student-t | 2.00 / 0.002 / <0.001  | 1.58 / 0.003 / <0.001  | 0.71 / 0.625 / 0.30 | 0.85 / 0.920 / <0.001 |
| GARCH-t   | 1.77 / <0.001 / 0.01   | 1.46 / <0.001 / 0.09   | 1.12 / 0.050 / 0.84 | 1.17 / 0.630 / 0.90   |
| FHS       | 1.23 / 0.172 / 1.00    | 1.20 / 0.188 / 0.90    | 0.92 / 0.345 / 0.87 | 0.85 / 0.968 / 0.99   |
| GARCH-EVT | 1.13 / 0.090 / 0.21    | 0.99 / 0.210 / 1.00    | 0.95 / 0.581 / 1.00 | 0.78 / 0.976 / 1.00   |

- Only FHS and GARCH-EVT pass the  $Z_2$  test at 5% and stay in the 90% MCS in all four markets; on Bitcoin every model stays in the MCS, so the comparison there is inconclusive

## Appendix: The Discovery Loop, Prompts and Checks

- Prompts (tools: [Semantic Scholar](#), [Elicit](#), [Google Scholar](#); check every DOI on [Crossref](#))
  - ▶ List peer-reviewed work since 2019 on conformal prediction under serial dependence, with DOI, exact guarantee and assumptions.
  - ▶ Give three rival explanations for crisis over-breaches of split conformal VaR and what each predicts for breach rates by regime.
  - ▶ Act as a hostile referee: list the strongest reasons this crisis effect is an artefact and the test that would settle each.
- Checks the researcher runs
  - ▶ **Look-ahead**: change the observations after day  $t - 1$  and confirm that  $\text{VaR}_t$ , the scale and the crisis flag do not move
  - ▶ **Dependent breaches**: Christoffersen and duration tests, or a block bootstrap of the hit sequence; **power**: 613 crisis days give 6.1 expected breaches of VaR 1%, and a true 2% rate is detected with probability 57%
  - ▶ **Multiplicity**: 4 markets  $\times$  2 levels  $\times$  3 methods = 24 tests: Holm (FWER) or BH (FDR); **misstated theorems**: e.g. “ACI guarantees conditional coverage”; **off-by-one quantiles**:  $k = \lceil (n + 1)(1 - \alpha) \rceil$  is not the default of `np.quantile`

## Appendix: Project Seed, Crisis-Conditional Coverage

- Question: can a distribution-free VaR 1% and ES 2.5% forecast achieve crisis-conditional coverage in all four markets under a pre-registered test?
- Replicate first: split conformal VaR 5% crisis rates (S&P 500 7.29%, EUR/RON 11.43%) and the ACI share of infinite VaR 1% (13.4%)
- Deliverables
  - ▶ A pre-registration (hypotheses, tests, corrections, power) dated before the first run
  - ▶ A candidate method, e.g. [Gibbs, Cherian & Candès \(2025\)](#) with the volatility regime as conditioning class; VaR tests by regime; ES tested with the ES regression backtest ([Bayer & Dimitriadis \(2022\)](#)) or  $Z_2$
  - ▶ Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log every prompt that shaped a design choice in AI\_USE.md and every error the assistant made in AI\_ERRORS.md: [github.com/danpele/MFM/tree/main/project](https://github.com/danpele/MFM/tree/main/project)

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